GRIDXRAY · BY ZYNPLEX

GridXray investigation report

Date: 2026-10-30 02:58:45 · ID 748c87f53678

Scenario

easy Network	**0** Seed	**22** Experiment budget
15 Secondary substation	**muse-spark-1.3** Models · Spark	**muse-wonder-1.0** Models · Wonder

Verified result

14/15 connections recovered out of the real ones · F1 93% · 22 experiments

14 Correct connections	**1** Wrong	**0/1** Wrong but flagged uncertain
2 Field incidents	**2/2** Detected by the AI	

Field incidents during the investigation

Faults and switching operations applied to the real network while the AI was working. The AI was never told: it had to notice them in its own readings.

0:39

- 🔥 Meter fault SS-01: meter offline (stopped reporting)

✓ Detected by the AI at 2:38, on experiment 1 after the change · 6 earlier experiments set aside

SS-01

Planning agent re-planned: added 10 experiment(s) · 20 experiment budget · recommended a field crew

SS-01 went silent indicating a field failure/network change that set aside 6 old experiments leaving 12 substations unconfirmed. With 0 budget left minus 1 for the crew visit, 10 extra experiments at the cap are needed to re-confirm suspect and uncertain sites.

New commitment published at the change:
`f158ce34438c9d2c141a582c36e993c386c38f43931166581a6bd10146a373bf`

3:00

- 🔥 Meter fault SS-06: meter offline (stopped reporting)

✓ Detected by the AI at 3:39, on experiment 1 after the change · 10 earlier experiments set aside

SS-06

Planning agent re-planned: added 2 experiment(s) · 22 experiment budget · recommended a field crew

SS-06 meter went offline indicating equipment failure requiring a crew visit, invalidating prior confirmation for 7 substations. With 6 budget left minus 1 for the crew, 5 remain to re-confirm 7 sites so 2 extra experiments are needed.

New commitment published at the change:
`f158ce34438c9d2c141a582c36e993c386c38f43931166581a6bd10146a373bf`

The result is scored against the network as it ended; the original seal still verifies the network at the start.

Initial plan and planning agent

The AI was not told what kind of plan this is (up to date, old or with errors): it received only the drawing, the date it was drawn and today's date, and had to judge it from the evidence.

with errors Chosen for this test	**2025-09-25** drawn on · today 2026-10-30	**1.1 years** age
11/15 Plan against the real network	**14/15** GridXray's result	**80%** Planner's trust in the plan

Budget: 10 · set by the user

All 15 drawn connections fit geography and voltage profile, ruling out gross drawing errors, while only 2 of 15 agree with the 24h correlation tree. Low correlation agreement is weak evidence against the plan because that tree is only about half right on multi-feeder networks like this one with 3 switches. The plan is only 1.1 years old with full metering and no missing nodes, so the fixed budget of 10 is enough to spot-check feeder boundaries rather than rebuild the network.

Trust the recent plan as the baseline and use the 10 fixed experiments to confirm feeder assignment and switch states. Prioritize the three feeder heads under SS-01 and the deep SS-07/SS-13/SS-10 chain where correlation disagreement concentrates.

Agent's conclusion

Accept infer_tree candidate: east branch SS-02 feeds SS-03/SS-05/SS-11, SS-05 feeds SS-07 and SS-10, SS-07->SS-13, SS-10->SS-14, SS-11->SS-15; west SS-04 fed from Substation feeds SS-01/SS-06, SS-06 feeds SS-08/SS-12, SS-08->SS-09. check_hypothesis valid with rms <0.00023, worst SS-12/SS-09 within noise.

Connections proposed by GridXray

SECONDARY SUBSTATION	FED FROM	VERDICT

SS-01	SS-04	correct · ?
SS-02	SS-01	correct · ?
SS-03	SS-02	correct
SS-04	Substation	wrong
SS-05	SS-02	correct
SS-06	SS-04	correct
SS-07	SS-05	correct
SS-08	SS-06	correct
SS-09	SS-08	correct
SS-10	SS-05	correct
SS-11	SS-02	correct
SS-12	SS-06	correct
SS-13	SS-07	correct · ?
SS-14	SS-10	correct
SS-15	SS-11	correct

Real connections not found: Substation–SS-01

Experiments run

#	ACTION	STRONGEST RESPONSES (MPU/MW)
1	1500.0 kW @ SS-14 · set aside: the network changed afterwards	SS-14 4.36, SS-10 3.95, SS-07 3.28, SS-13 3.24
2	1500.0 kW @ SS-09 · set aside: the network changed afterwards	SS-09 3.70, SS-08 3.30, SS-06 3.00, SS-12 2.99
3	1500.0 kW @ SS-02 · set aside: the network changed afterwards	SS-07 3.08, SS-15 3.05, SS-02 3.04, SS-03 3.02
4	1500.0 kW @ SS-04 · set aside: the network changed afterwards	SS-09 2.55, SS-12 2.52, SS-08 2.50, SS-06 2.48
5	1500.0 kW @ SS-07 · set aside: the network changed afterwards	SS-13 3.65, SS-07 3.62, SS-05 3.27, SS-14 3.26
7	1500.0 kW @ SS-03 · set aside: the network changed afterwards	SS-03 3.61, SS-05 3.06, SS-14 3.06, SS-02 3.04
8	1500.0 kW @ SS-15 · set aside: the network changed afterwards	SS-15 4.21, SS-11 3.40, SS-03 3.06, SS-13 3.04
9	1500.0 kW @ SS-11 · set aside: the network changed afterwards	SS-11 3.39, SS-15 3.39, SS-07 3.08, SS-10 3.05
10	1500.0 kW @ SS-05 · set aside: the network changed afterwards	SS-07 3.23, SS-10 3.23, SS-05 3.23, SS-14 3.22
11	1500.0 kW @ SS-14	SS-14 4.27, SS-10 3.94, SS-05 3.26, SS-07 3.25
12	1500.0 kW @ SS-07	SS-13 3.64, SS-07 3.57, SS-14 3.24, SS-05 3.24
13	1500.0 kW @ SS-02	SS-15 3.08, SS-05 3.07, SS-11 3.03, SS-03 3.01
14	1500.0 kW @ SS-10	SS-14 3.93, SS-10 3.92, SS-05 3.27, SS-13 3.25
16	1500.0 kW @ SS-12	SS-12 3.76, SS-08 2.96, SS-09 2.91, SS-04 2.46
17	1500.0 kW @ SS-08	SS-09 3.33, SS-08 3.26, SS-12 2.95, SS-04 2.42
18	1500.0 kW @ SS-09	SS-09 3.68, SS-08 3.30, SS-12 2.95, SS-04 2.52
19	1500.0 kW @ SS-04	SS-09 2.48, SS-04 2.48, SS-12 2.45, SS-08 2.44

20	1500.0 kW @ SS-05	SS-13 3.28, SS-05 3.26, SS-07 3.25, SS-10 3.21
21	1500.0 kW @ SS-11	SS-15 3.36, SS-11 3.35, SS-03 3.06, SS-13 3.05
22	1500.0 kW @ SS-15	SS-15 4.26, SS-11 3.36, SS-10 3.06, SS-05 3.05

Hypotheses

HYPOTHESES	CONFIDENCE
Baseline is documented plan (80% trust): SS-01 fed from Substation with three feeder heads SS-02/SS-03/SS-04. East chain SS-02-SS-05-SS-07-SS-13-SS-10-SS-14(+SS-15) and west chain SS-04-SS-06-SS-08-SS-09 disputed vs geography putting SS-11 between SS-01 and SS-02/SS-03. First probe deep ends (SS-14, SS-13, SS-09) plus heads (SS-02, SS-04) to reveal shared paths.	80%
SS-14 under SS-10; SS-13 under SS-07; SS-05 splits to SS-07 and SS-10; SS-15 is upstream of SS-05 not under SS-10; SS-09 under SS-08 under SS-06 chain under SS-04; east root order SS-02/SS-03/SS-11/SS-15 still ambiguous. Probe SS-03, SS-15, SS-11, SS-05.	70%
FIELD CHANGE: SS-01 meter went offline after exp6; first 6 experiments set aside. Only post-change exps (SS-03, SS-15, SS-11, SS-05) valid. Sending crew to SS-01 to check switch/recloser and meter damage before re-testing.	60%
After 2nd field change (SS-06 meter burned, SS-01 offline) only exps 11-14 valid. East hub confirmed: SS-05 feeds SS-07->SS-13 and SS-10->SS-14, all under SS-02 head. West/root chain (SS-04/SS-12/SS-06/SS-08/SS-09 and SS-01/SS-11/SS-15/SS-03) still ambiguous. Re-testing west + root next.	65%

Muse Wonder reviews

midpoint review

Not justified

- Candidate inverts evidence: SS-02 parent SS-03, and SS-03 parent SS-12/SS-04-chain. Perturbing SS-02 makes SS-03 drop equally (SS-03 downstream), while perturbing SS-04 does not affect SS-02/SS-03. That chain is impossible, not merely inferred.
- SS-13 parent SS-05 contradicts SS-07 perturbation: only SS-13 drops equally, so SS-13 is below SS-07 (documented), not a sibling under SS-05.
- Direct parents claimed “measured” for SS-05 ← SS-15, SS-01 ← Substation, SS-04 ← Substation, SS-14 ← SS-10 are unproven: SS-15, SS-01, SS-10 never perturbed; only ancestor/descendant sets and depth order are known. Likewise SS-11 ← SS-03 and SS-10 ← SS-15 are guesses; SS-11, SS-10, SS-15, SS-03 unprobed. West direct links SS-06–SS-08–SS-09 also only show nested descent from SS-04.

Unconsidered

Outdated switch state: recloser 1 at SS-01 or sectionalizer 0 at SS-10 changed after the 2025 plan, re-feeding the eastern cluster; swapped/faulty SS-02/SS-03 meters; unmetered intermediate junction under SS-02 feeding SS-03/SS-11/east.

Plan (budget 4)

1. `perturb load SS-03, 1500 kW, 1` : orders the east root and kills the SS-02 inversion.
2. `perturb load SS-15, 1500 kW, 1` : separates SS-15 above SS-05/SS-10 versus documented below SS-10.
3. `perturb_load SS-01, 1500 kW, 1` : tests whether SS-02 and/or SS-04 sit under SS-01 or Substation. Reserve one for SS-05 follow-up.

Usage and cost

MODEL	TOKENS (IN / OUT / REASONING)	COST (USD)
muse-spark-1.3	138,411 / 44,117 / 40,003	$0.263
muse-image-1.0	0 / 0 / 0	$0.020
sam-3.1	0 / 0 / 0	$0.020
muse-wonder-1.0	3,150 / 17,330 / 16,241	$0.359
Total	**203,008**	**$0.662**

Sealed topology

Published before the AI saw anything: SHA-256(secret + true topology).
`f158ce34438c9d2c141a582c36e993c386c38f43931166581a6bd10146a373bf`
Verified: the real network matches the seal.

Tools the AI used

check_hypothesis, consensus, dispatch_crew, geographic_prior, get_evidence, get_overview, infer_tree, note_hypothesis, passive_correlation, run_experiments, submit_result, suggest_experiments

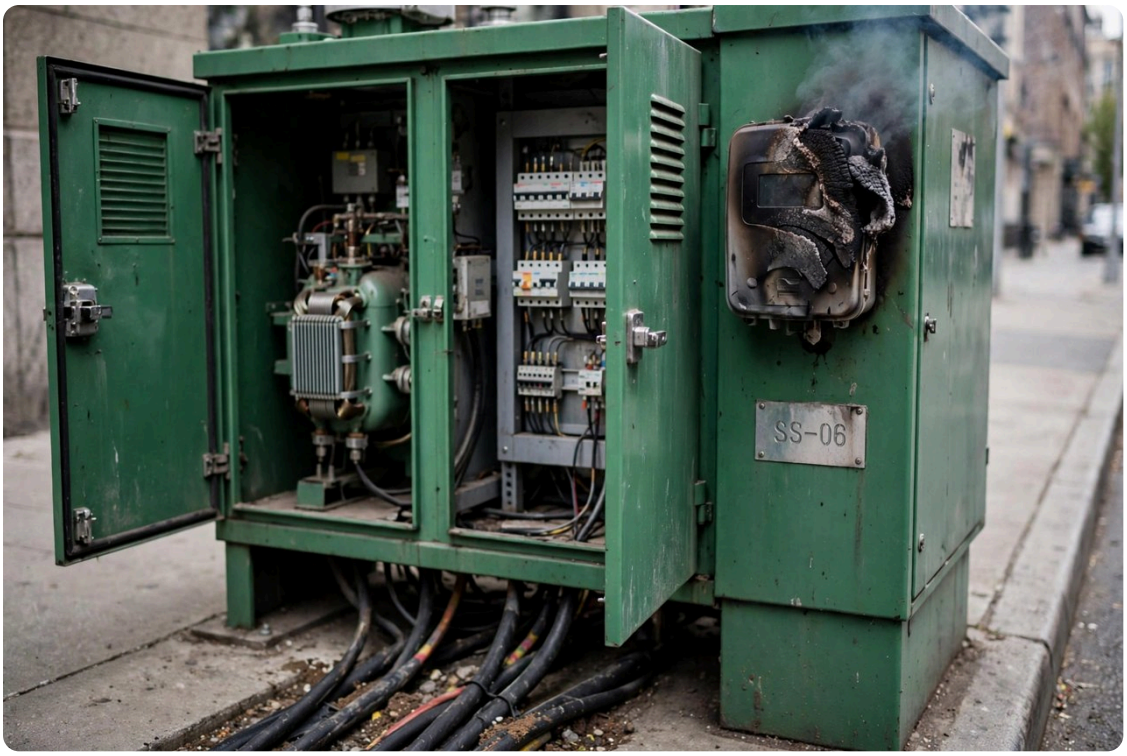
Field crew visits

Photo drawn by Muse Image from what is physically at the site, segmented by SAM 3.1 and read by Muse Spark. The investigator only received the reading.



SS-10 · SW-0 closed · matches the records

Switch SW-0 shows I on red indicating closed at kiosk site SS-10 with underground cables.



SS-06 · unknown · no switch on record here

🔥 **FAULT SEEN: the smart meter at SS-06 is damaged**

Green street kiosk SS-06 with underground cables shows scorched smoking smart meter and no readable switch indicator.

How this was scored

An independent checker with no AI compares every proposed connection with the simulator's ground truth. The AI never saw the true topology: only equipment, records, meter readings and the results of its own experiments.

Reproduce

```
uv run gridxray investigate --level easy --seed 0 --budget 22
```